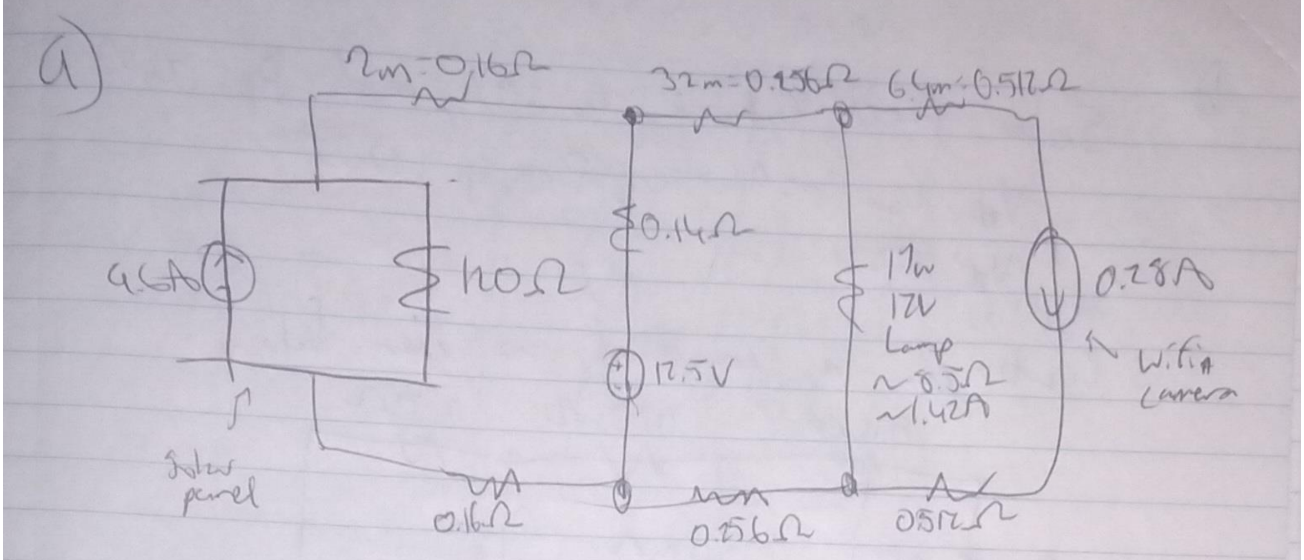
(a) Draw a full circuit diagram of the system described above, correctly annotating **all** circuit components with the values you have been assigned. Where possible, simplify the diagram by combining series and parallel resistances and sources within a loop. Do not apply Thévenin modelling yet.

(2 marks)



* Wires have 0.08 Ω of resistance/m.
* No series or parallel connections between resistors or sources.

* Bulb can also be modelled as resistor using equation $\frac{V^2}{P} = R \Rightarrow \frac{12^2}{17} = \sim 8.5\Omega$ and draws current $\Rightarrow \frac{P}{V} = i \Rightarrow \frac{17}{12} \approx 1.42A$

(b) The solar panel measurements give parameters which directly apply to a Norton model. However, a Thévenin model will simplify the analysis process. Perform a source transformation on the solar panel and redraw the full circuit, including the Thévenin equivalent model.

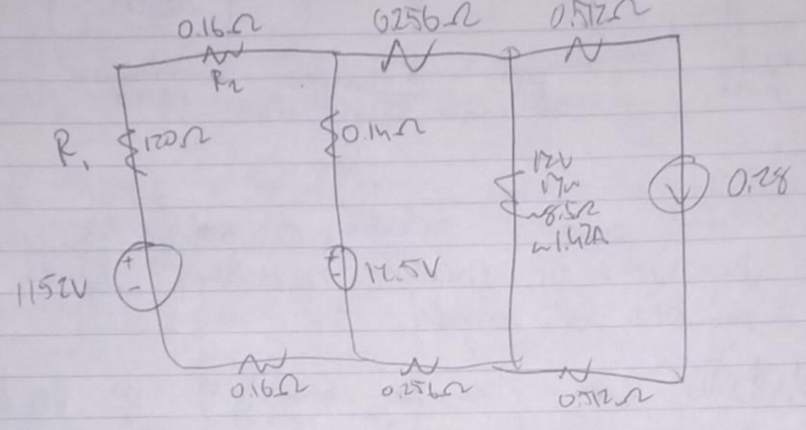
(3 marks)

b. Solar panel exhibits $i_{sc} = 9.6$, and $R_p = 120\Omega$

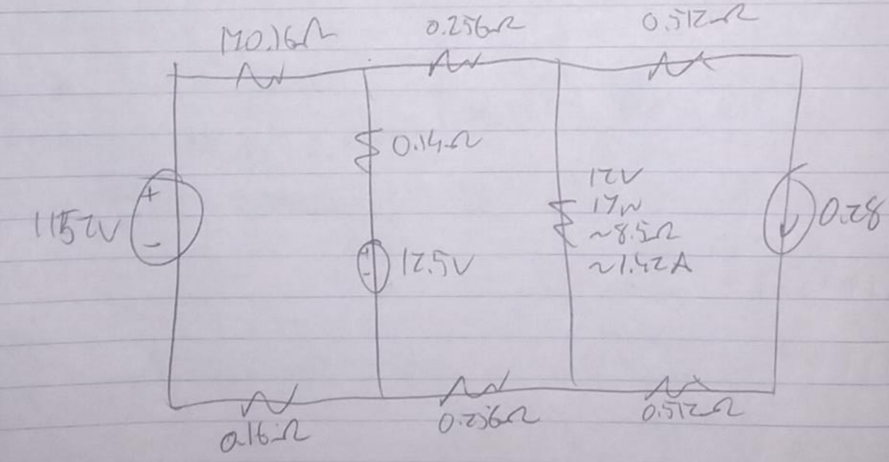
$\therefore V_{Th} = i_{sc} \cdot R_p = 9.6 \times 120$, $R_{Th} = R_p = 120\Omega$

$V_{Th} = 1152V$

$\therefore$ Circuit can be modelled as seen below:



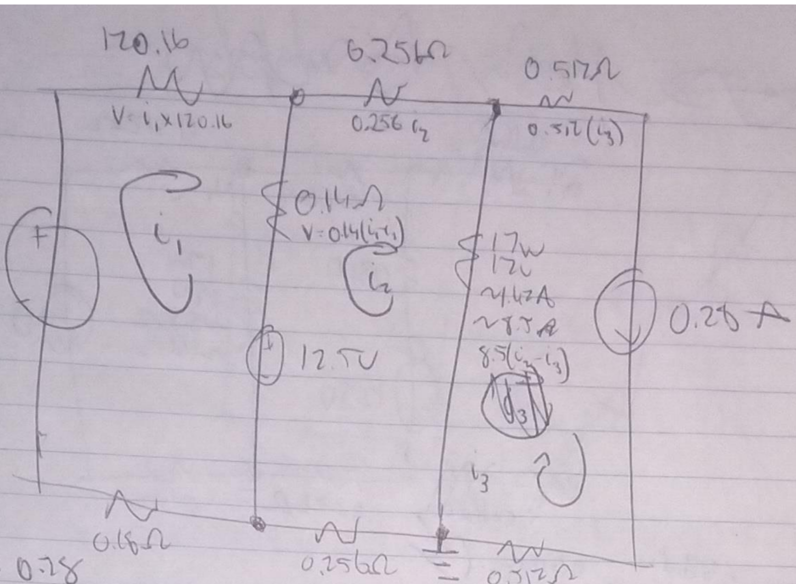
* R_1 and R_2 are in series and can therefore be simplified by sum $\Rightarrow 120 + 0.16 = 120.16\Omega$



(c) Using mesh analysis, calculate the mesh currents at midday, when the solar panel delivers its rated current (a) and the light bulb and camera are drawing current. Show all working.

(4 marks)

(c)



$i_3 = 0.28$

① $1152 - 170.16 i_1 - 0.14(i_1 - i_2) - 12.5 - 0.16 i_1 = 0$

② $12.5 + 0.14(i_1 - i_2) - 0.256 i_2 - 8.5(i_2 - i_3) - 0.256 i_2 = 0$

① $1152 - 12.5 = 170.16 i_1 + 0.14(i_1 - i_2) + 0.16 i_1$

$1139.5 = 170.32 i_1 + 0.14 i_1 - 0.14 i_2$

$1139.5 = 170.46 i_1 - 0.14 i_2$

② $12.5 = 0.256 i_2 + 8.5(i_2 - 0.28) + 0.256 i_2 - 0.14(i_1 - i_2)$

$12.5 = 9.152 i_2 - 0.14 i_1 - 2.38$

$14.88 = 9.152 i_2 - 0.14 i_1$

Solving simultaneous:

① $1139.5 = 170.46 i_1 - 0.14 i_2$

$14.88 = 9.152 i_2 - 0.14 i_1$

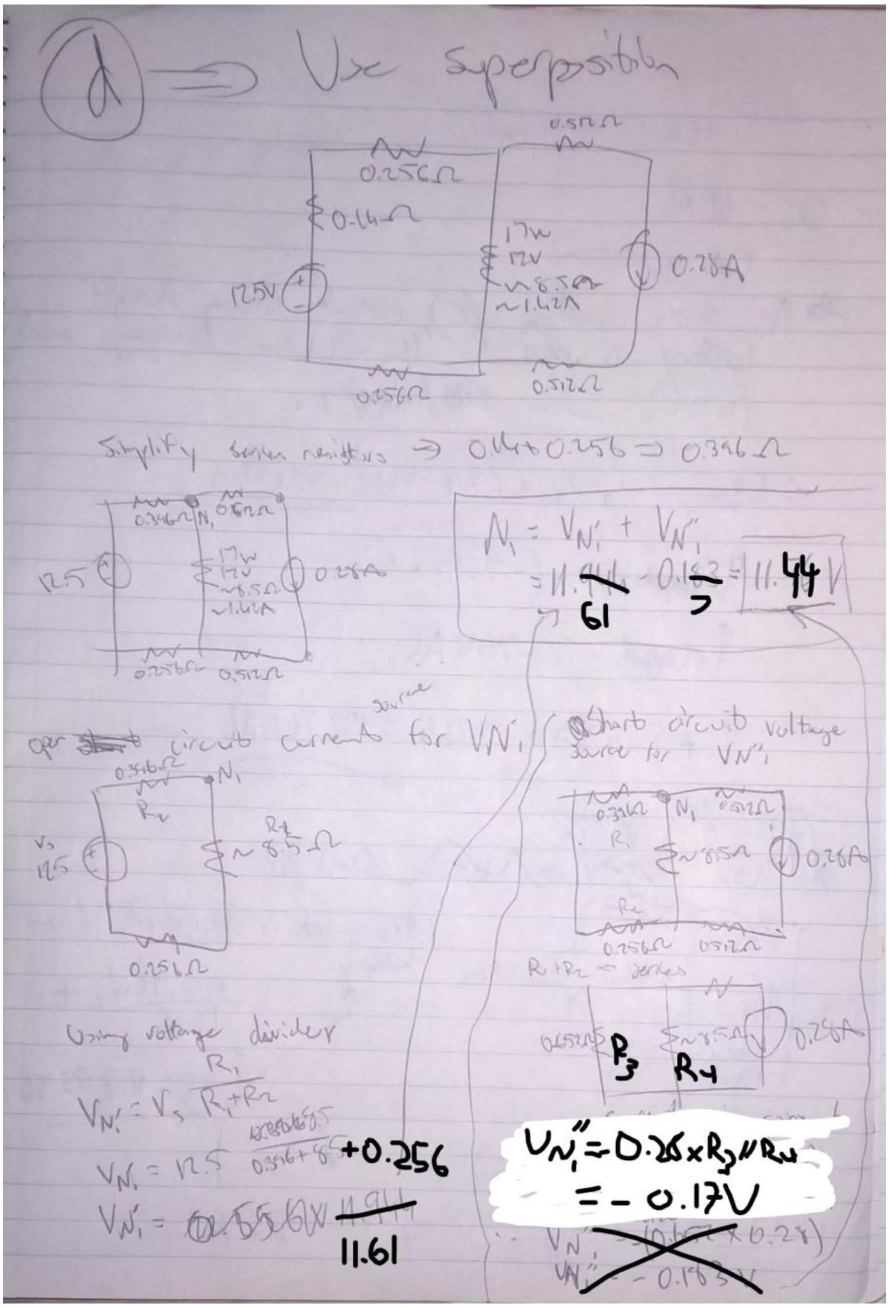
Solving in calculator $\Rightarrow i_1 = 9.462\text{A}$, $i_2 = 1.771\text{A}$

$i_3 = 0.28\text{A}$

Daytime i_1 = 9.462A, Daytime i_2 = 1.771A, Daytime i_3 = 0.28A

(d) At night, the incubator light and camera continue operate using only battery power. During this time, the solar panel (and its associated resistor) acts as an open circuit. Draw the updated circuit and **use superposition** to find the voltage at N1 with respect to ground. **Note – be careful about the polarity of the voltage drops across components.**

(4 marks)



N1’ = 11.61V, N1’’ = -0.17V, N1 = 11.44V

(e) Use circuit simulation software to verify your calculations in (c) and (d). Capture screenshots of the software showing your built circuit for the mesh analysis and both superposition sub-circuits and report the values given. Use the table provided to compare your calculated and simulated answers. The simulated values (those in the provided table) must be shown in the simulation screenshots.

(3 marks)

Figure 1: Circuit for mesh analysis

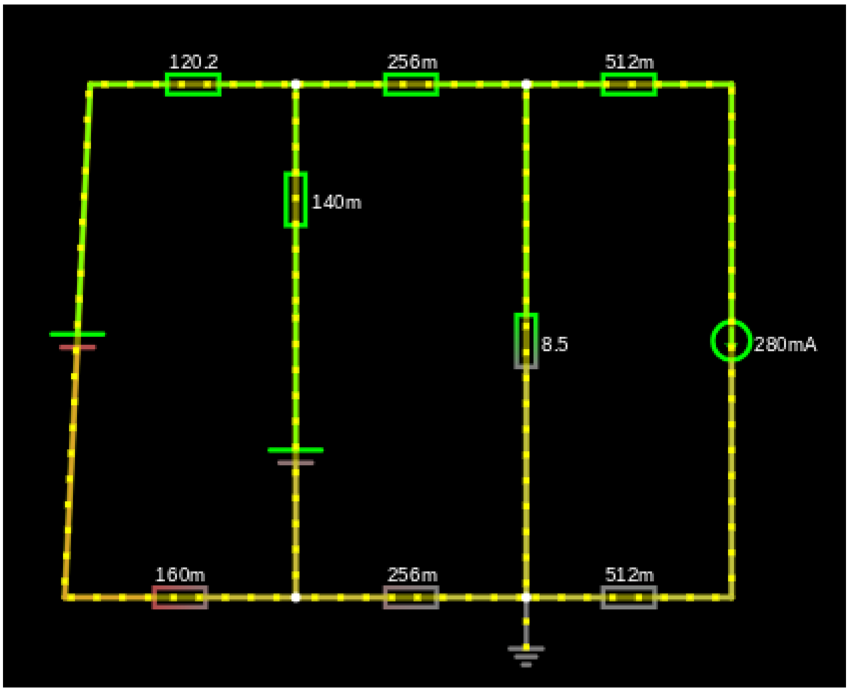
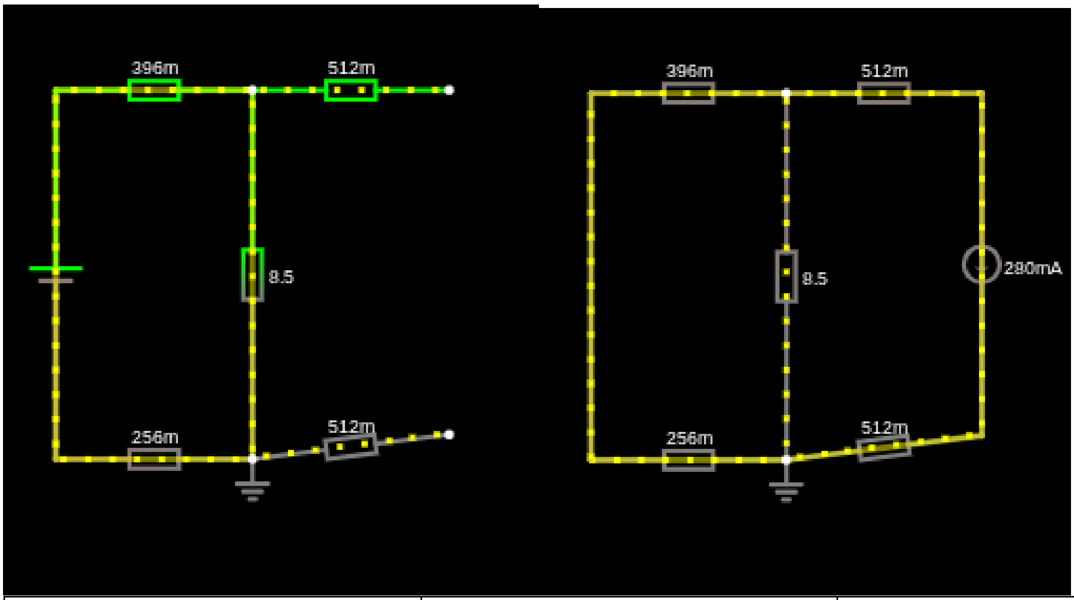


Figure 2: Superposition Circuits: Open Circuit Current Source (left) and Short Circuit Voltage Source (right).



Values	Calculated	Simulated
I1	9.462A	9.462A
I2	1.771A	1.771A
I3	0.28A	0.28A
N1’	11.61V	11.609V

N1''	-0.17V	-0.170V
N1	11.44V	11.44V

(f) "Assume it is a sunny winter day and the solar panel provides its rated midday current (a) for 4 hours. The 120 Ah battery starts the day at 50% capacity (60 Ah stored). While the battery is charging, the light bulb and WiFi camera are still running. How much current flows through the battery and how much charge (in Amp-hours) will the battery store after 4 hours of charging? After this 4-hour charge, if the next days are cloudy (i.e. the solar panel current source and parallel resistance act as an open circuit), how long will it take for the battery to go flat and for all of the devices to stop working? Perform the calculations below.

(4 marks)

$i = 96A$ $t = 4$ hours

(e)

$Q_i = 60Ah$

First question

As seen within (c), current flowing through battery is equal to $(i_1 - i_2)$ when charging

$$\therefore I_{battery} = i_1 - i_2 = \frac{9.462 - 1.771}{} = 7.691\ A$$

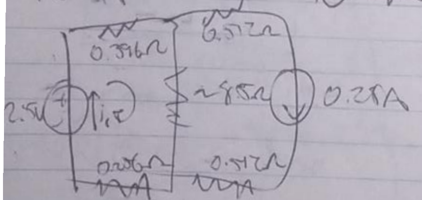
$$q_{charged} = \frac{7.691 * 4}{}$$

$$q_{charged} = 30.764\ Ah$$

$$q_i + q = 60 + 30.764 = 90.764\ Ah$$

~~Second question~~

~~As seen in mesh analysis~~



1.626A is dissipated from battery

$$\therefore \frac{90.764}{1.626} = t$$

$$\therefore t = \cancel{56}\ 55.8\ hours$$

Using mesh analysis

$i_2 = 0.28$

$$12.5 - 0.396 i_1 - 0.256 i_1 - 8.5(i_1 - 0.28) = 0$$

$$\frac{12.5 + 8.5 \times 0.28}{9.152} = \frac{9.152 i_1}{9.152} \Rightarrow i_1 = 1.626\ A$$

Charge Stored = 90.764Ah
Discharge time = 55.8 hours

Total: (20 marks)